

NEWTONIAN MECHANICS • PRESSURE

Why your ears hurt at the bottom of a deep pool

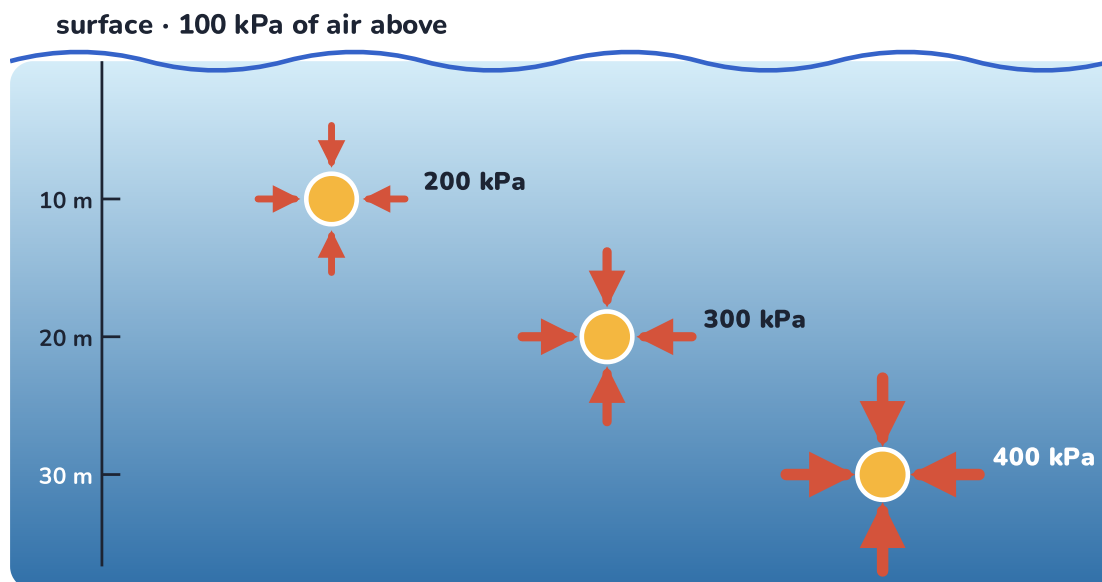
A stiletto heel can dent a floor that an elephant walks across safely. Car brakes turn a light push of your foot into a force that stops a tonne of metal. This chapter explains both, with one idea.

Sec 3–4 • Express

Exam weight: high

5 worked examples

11 practice questions



SEAB 6091 LEARNING OUTCOMES • BY THE END YOU CAN

1. define the term **pressure** in terms of force and area
2. recall and apply $p = F / A$ to new situations and problems
3. describe and explain the **transmission of pressure in hydraulic systems**, with particular reference to the hydraulic press
4. recall and apply $p = h\rho g$ for the pressure due to a liquid column
5. describe how the **height of a liquid column** may be used to measure atmospheric pressure
6. describe the use of a **manometer** to measure pressure difference

Name _____

Class _____

Date _____

Predict 1

Answer in the Annex.
Guess first!

? Define
pressure.

? What is 1 Pa?

? Why must the
force be *normal* to
the surface?

? How many m^2 is
1 cm^2 ?

Pressure in solids

heading

Same force, different area, very different effect.

predict

PREDICT FIRST · NO CALCULATOR, JUST YOUR GUT

Who would you rather be stepped on by?

A 60 kg person in stiletto heels, standing on one heel (heel area about 1 cm^2). Or a 5000 kg elephant standing on four feet (each foot about 0.1 m^2).

- ☐ A The person in stilettos hurts more
- ☐ B The elephant hurts more
- ☐ C They hurt about the same

Why do you think so?

concept

Pressure is the force acting normally (at right angles) per unit area of a surface.

The SI unit of pressure is the **pascal (Pa)**. One pascal is a force of one newton spread over one square metre, so $1 \text{ Pa} = 1 \text{ N/m}^2$. That is tiny: a sheet of paper lying on a desk exerts about 1 Pa. So you will mostly meet **kPa** ($\times 10^3$) and sometimes **MPa** ($\times 10^6$).

formula

$$p = F / A$$

p	pressure	Pa or N/m^2
F	force acting normally on the surface	N
A	area the force acts on	m^2

Rearrange as needed: $F = pA$, $A = F / p$. Convert cm^2 to m^2 first: $1 \text{ cm}^2 = 1 \times 10^{-4} \text{ m}^2$.

examiner

EXAMINER'S TIP · THE MOST COMMON LOST MARK IN THIS CHAPTER

1 m = 100 cm, but **1 m^2 = 10 000 cm^2** . Students divide by 100 instead of 10 000 and get an answer 100 times too big.

Safer habit: convert each length to metres *before* you multiply. $20 \text{ cm} \times 10 \text{ cm}$ becomes $0.20 \text{ m} \times 0.10 \text{ m} = 0.020 \text{ m}^2$.

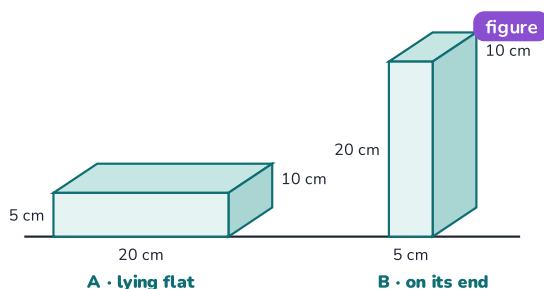
? Give two examples where we want high pressure, and two where we want low pressure.

We want HIGH pressure (small area)

- Knife blades and scissors cut with a thin edge
- Nails and drawing pins have sharp points
- Injection needles pierce skin easily
- Ice-skate blades press hard on the ice

We want LOW pressure (large area)

- Tractors and tanks use wide tyres or tracks so they don't sink into mud
- Snowshoes and skis spread your weight on snow
- Building foundations are wider than the walls above them
- School bags have wide padded straps



The same block rests on two different faces. Its weight is 20 N either way.

DID YOU KNOW?

$\approx 3 \times$

A camel's wide padded feet let it walk on soft sand. A horse of the same weight sinks in, because its hooves press on about a third of the area.

Worked example 1

[3]

Try it in the box. Full solution in the Annex.

The block above weighs 20 N and measures 20 cm $\times$ 10 cm $\times$ 5 cm. Calculate the **greatest** and the **least** pressure it can exert on the ground.

Q1

● ○ ○

Warm-up

[2]

A student of weight 600 N stands on one foot. The area of the sole touching the floor is 150 cm². Calculate the pressure on the floor. Give your answer in kPa.

Q2



Exam

[4]

A tractor of mass 3000 kg has four tyres. Each tyre touches the ground over an area of 0.15 m^2 . Take $g = 10 \text{ N/kg}$.

question

(a) Calculate the pressure the tractor exerts on the ground.

[2]

(b) Explain why tractors used on muddy farms have very wide tyres.

[2]

Pressure in liquids

heading

Why deep water squeezes you, whatever shape the container is.

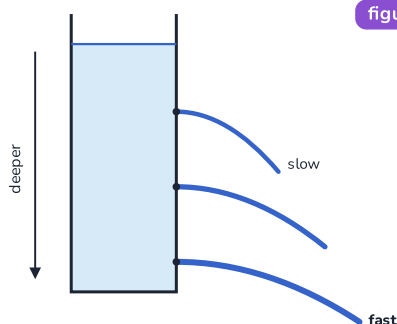
? List four facts about pressure in a liquid.

? Does a wider tank give more pressure at the bottom?

concept

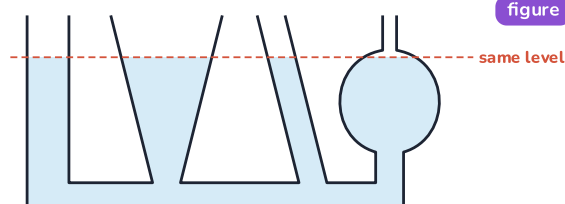
The liquid above any point has weight. That weight pushes on everything below it. Pressure in a liquid at rest has four properties.

- It acts **in all directions** at any point: up, down and sideways.
- It **increases with depth**, because there is more liquid above pushing down.
- It **increases with density**. A denser liquid has more weight in the same height.
- It **does not depend on the shape or width** of the container, only on the vertical depth.



figure

Water leaves the **deepest hole fastest** because the pressure there is greatest.

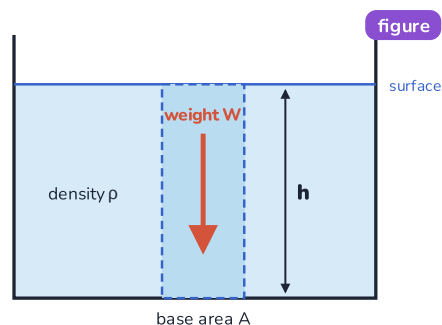


row figure

Connected vessels of any shape fill to the **same level**. If one level were higher, the pressure at the bottom would be bigger on that side and push the liquid back until the levels matched.

? Derive $p = h\rho g$ in four lines.

? What does h measure, exactly?



Imagine a column of liquid of height h standing on area A .

Where $p = h\rho g$ comes from

row
concept

Volume of the column: $V = Ah$

Mass of the column: $m = \rho V = \rho Ah$

Weight of the column: $W = mg = \rho Ahg$

Pressure on the base: $p = W / A = \rho Ahg / A = h\rho g$

$A = h\rho g$

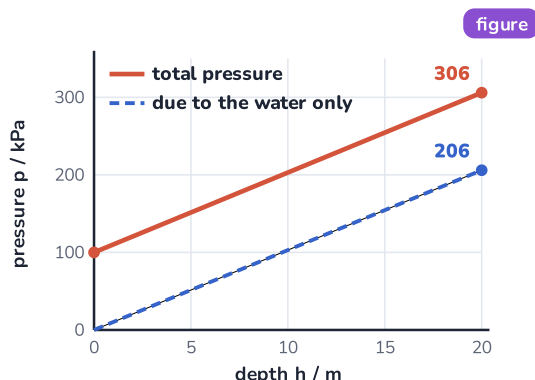
The area A cancels out. That is why the width of the container does not matter.

$$p = h\rho g$$

p	pressure due to the liquid column	Pa
h	vertical depth below the surface	m
ρ	density of the liquid	kg/m^3
g	gravitational field strength	10 N/kg

formula

This is the pressure due to the liquid only. Pressure at a depth in an open container is $p_{\text{total}} = p_{\text{atm}} + h\rho g$.



figure

EXAMINER'S TIP

row
examiner

"Pressure **due to the water**" means $h\rho g$ only.

"**Total** pressure" or "pressure **at** the diver" means $h\rho g$ + atmospheric pressure.

Underline these words in the question before you calculate.

Seawater, $\rho = 1030 \text{ kg/m}^3$. Both lines are straight because p is directly proportional to h . The gap between them is always 100 kPa of air pressure.

Worked example 2

[3]

Try it in the box. Full solution in the Annex.

A diver swims 20 m below the surface of the sea. The density of seawater is 1030 kg/m^3 and atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$. Calculate the **total** pressure on the diver.

worked-example



Someone else's
working. Find the slip.

Over 1100 x

At the bottom of the Mariana Trench, about 11 000 m down, the water pressure is roughly $11\,000 \times 1030 \times 10 \approx 1.1 \times 10^8$ Pa. That is more than a thousand times the air pressure on you right now. Every 10 m of water adds about one extra atmosphere.

A student calculated the pressure due to the water 50 cm below the surface of a pond ($\rho = 1000 \text{ kg/m}^3$).

L1 $p = h\rho g$

L2 $p = 50 \times 1000 \times 10$

L3 $p = 500\,000\text{ Pa}$

Circle the line with the mistake. Then write the correct working below.

A water tank is filled to a depth of 3.0 m. The density of water is 1000 kg/m^3 .

(a) Calculate the pressure on the base due to the water.

(b) The base has an area of 2.0 m^2 . Calculate the force of the water on the base.

(c) Explain why water comes out of a tap at the bottom of the tank faster than from a tap near the top.

Q4



Stretch

[3]

A tall thin vase and a wide bowl are both filled with water to a depth of 12 cm. Your friend says the bowl holds more water, so the pressure on its base must be bigger. Is your friend right? Explain, and state what **is** bigger for the bowl.

Q5



Stretch

[2]

Dams are built much thicker at the bottom than at the top. Explain why, using ideas about pressure.

03

? What happens to pressure applied to an enclosed liquid?

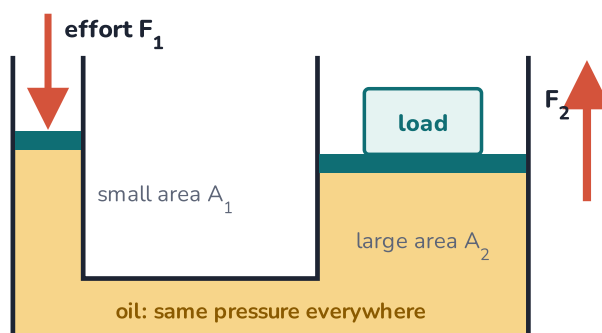
? Why must a hydraulic liquid have no air bubbles?

Hydraulic systems

How a push from your foot stops a car.

Liquids are almost **incompressible**. When you push on a liquid trapped in a closed system, the pressure you add is **transmitted equally to every part of the liquid**.

A hydraulic press connects a **small piston** to a **large piston** through a pipe of oil. A small effort on the small piston makes a pressure $p = F_1 / A_1$. The same pressure acts on the large piston, which has a much larger area, so it produces a much larger force $F_2 = p \times A_2$.



The pressure is the same on both pistons. The force is bigger where the area is bigger.

? Write the hydraulic press equation.

? If the force is multiplied 50 times, what happens to the distance moved?

Worked example 3

[5]

Try it in the box. Full solution in the Annex.

Error hunt 2

Someone else's working. Find the slip.

$$F_1 / A_1 = F_2 / A_2$$

formula

F_1	effort on the small piston	N
A_1	area of the small piston	m ² or cm ²
F_2	force on the large piston (the load)	N
A_2	area of the large piston	same unit as A_1

The force is multiplied by A_2 / A_1 . Any area unit works here, as long as both areas use the same one, because only the ratio matters.

Nothing for free

concept

The oil pushed out of the small cylinder fills the large cylinder, so the volumes match. $A_1 d_1 = A_2 d_2$. The large piston moves a much shorter distance than the small one. You gain force but lose distance, so no energy is created.

In a hydraulic press, the small piston has an area of 4.0 cm² and the large piston has an area of 200 cm². An effort of 50 N is applied to the small piston. Calculate (a) the pressure in the oil, (b) the force on the large piston, and (c) how far the large piston rises when the small piston is pushed down 10 cm.

worked-example

Effort 20 N on a piston of area 2 cm². The large piston has area 50 cm². A student finds the output force F_2 .

error-hunt

L1 $F_1 / A_1 = F_2 / A_2$

L2 $20 / 50 = F_2 / 2$

L3 $F_2 = 0.8 \text{ N}$

Circle the line with the mistake. Then write the correct working below.

Q6



Exam

[4]

A car's brake pedal pushes a piston into a cylinder of brake fluid. Pipes carry the fluid to wider pistons at each wheel, which press the brake pads onto the wheels.

question

- (a) Explain why the force on each brake pad is larger than the force on the pedal piston.

[2]

- (b) A mechanic warns that an air bubble in the brake fluid is dangerous. Explain why.

[2]

04

? What causes atmospheric pressure?

? Roughly how big is it at sea level?

Activity 1

Atmospheric pressure and the barometer

heading

We live at the bottom of an ocean of air.

Air has weight. The column of air above you, about 100 km tall, pushes on every surface. At sea level atmospheric pressure is about 1.0×10^5 Pa (100 kPa). That is like a 1 kg mass resting on every square centimetre of your skin. You don't feel it because the fluids in your body push outwards just as hard.

concept

Atmospheric pressure **falls as you go higher**, because there is less air above you. On the summit of Mount Everest it is only about a third of its sea-level value.

activity

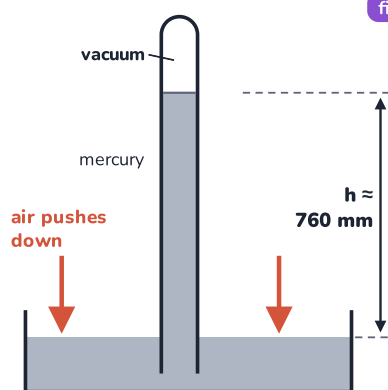
TRY IT AT HOME

The card that holds up water

You need: a plastic cup, a postcard or stiff card, water, a sink

1. Fill the cup right to the brim with water.
2. Slide the card over the top so no air is trapped.
3. Hold the card, turn the cup upside down over the sink, then let go of the card.

The card stays on and the water stays in. Explain why, using atmospheric pressure.



A simple mercury barometer.

figure

Measuring it with a liquid column

Fill a 1 m glass tube with mercury, cover the end, and turn it upside down into a dish of mercury. The mercury falls until the column is about **760 mm** tall, leaving a **vacuum** above it.

The air pushing on the dish holds the column up. At the level of the dish surface, **atmospheric pressure = pressure due to the mercury column**:

$$p_{\text{atm}} = h\rho g$$

So "760 mm Hg" (or 76 cm Hg) is a way of stating atmospheric pressure.

? Does tilting the tube change h ?

? Why mercury and not water?

What changes h , and what doesn't

- Tilting the tube, or using a wider or narrower tube, does **not** change the **vertical** height h . The mercury just fills more of the tilted tube.
- Pushing the tube deeper into the dish does **not** change h either.
- h **does** change when atmospheric pressure changes: lower on a mountain, a little lower before a storm.
- Mercury is used because it is very dense ($13\,600\text{ kg/m}^3$). A water barometer would need a tube over **10 m** tall.
- Air getting into the vacuum at the top would push down on the column and make h **smaller** than it should be.

EXAMINER'S TIP

Always measure h **vertically**, from the mercury surface in the dish to the top of the column. Measuring along a tilted tube is a very common slip.

"Explain why the column stays up" needs the word **atmospheric pressure acting on the surface of the mercury in the dish**. Just saying "the air holds it up" usually earns no mark.

Worked example 4

[2]

Try it in the box. Full solution in the Annex.

On a mountain top the mercury column in a barometer is 60 cm tall. The density of mercury is $13\,600\text{ kg/m}^3$. Calculate the atmospheric pressure there in pascals.



Multiple choice. Tick one box.

A mercury barometer reads 760 mm. The tube is tilted so that it makes an angle of 60° with the horizontal. What happens to the mercury?

A The vertical height of the column falls below 760 mm.

B The vertical height of the column rises above 760 mm.

C The vertical height stays 760 mm, but the mercury fills a longer length of tube.

D The mercury runs out of the tube into the dish.

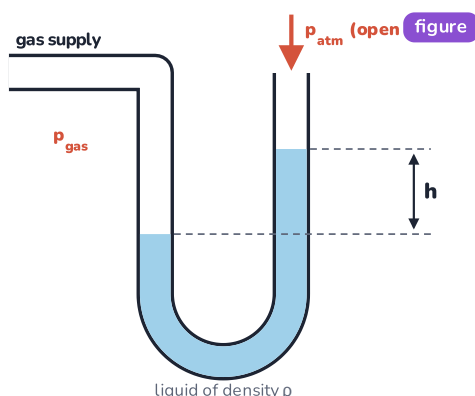
? What does a manometer measure?

? How do you tell if the gas pressure is above or below atmospheric?

heading

Measuring a pressure difference with a U-tube.

A **manometer** is a U-shaped tube containing a liquid (often water or mercury). One arm connects to the gas supply, and the other is open to the air. If the two liquid levels are equal, the gas pressure equals atmospheric pressure. If the gas pushes the liquid **down** on its side, the gas pressure is **greater** than atmospheric. The difference in levels h measures the pressure difference:



Gas pressure is higher than atmospheric here, so the liquid is pushed round into the open arm.

$$p_{\text{gas}} = p_{\text{atm}} + h\rho g$$

h vertical difference between the two levels

ρ density of the manometer liquid kg/m^3

If the liquid is higher on the gas side, the gas pressure is below atmospheric: $p_{\text{gas}} = p_{\text{atm}} - h\rho g$. The width of the tube does not change h .

[3]

Try it in the box. Full solution in the Annex.

A water manometer is connected to a gas supply. The water in the open arm stands 25 cm higher than the water in the arm connected to the gas. Atmospheric pressure is 1.0×10^5 Pa and the density of water is 1000 kg/m^3 . Calculate the pressure of the gas.

Multiple choice. Tick one box.

Q9

Exam

The U-tube of a manometer is replaced by one with twice the cross-sectional area. It is connected to the same gas supply. What happens to the difference in liquid levels, h ?

B It stays the same.

D It becomes four times as large.

A mercury manometer is connected to a gas cylinder. The mercury in the arm connected to the gas is 4.0 cm **higher** than in the open arm. Atmospheric pressure is 76 cm Hg. The density of mercury is 13 600 kg/m³.

(a) State whether the gas pressure is above or below atmospheric pressure. Explain how you know.

(b) Calculate the gas pressure in cm Hg, and then in Pa.

Pull it together

Check yourself before you move on.

KEY WORDS

pressure

force acting normally per unit area

pascal (Pa)

1 N acting on 1 m²

depth

vertical distance below the liquid surface

incompressible

cannot be squashed into a smaller volume

atmospheric pressure

pressure of the air, about $1.0 \times 10^5 \text{ Pa}$ at sea level

manometer

U-tube that measures a pressure difference

Q10

Multiple choice. Tick one box.

[1]

Which unit is the same as the pascal?

B N/m^2

D N/m

Q11

Multiple choice. Tick one box.

[1]

Two tanks are filled to the same depth. Tank X holds water (1000 kg/m^3). Tank Y holds oil (800 kg/m^3) and is three times wider. Which statement about the pressure on the base due to the liquid is correct?

- ☐ **A** It is the same in both tanks.
- ☐ **B** It is greater in X.
- ☐ **C** It is greater in Y, because Y holds more liquid.
- ☐ **D** It is greater in Y, because Y has a larger base.

mcq

summary

SUMMARY · EXPLAIN PRESSURE TO A SEC 1 STUDENT IN THREE SENTENCES

HOW CONFIDENT ARE YOU? TICK ONE, THEN DIARY THE REVIEW DATE

confidence

☐ **Shaky**

Redo the cue questions tomorrow.

Review on ____ / ____

☐ **Getting there**

Redo the worked examples in 3 days.

Review on ____ / ____

☐ **Nailed it**

Cue questions only, in a week.

Review on ____ / ____

A

Annex • Solutions

Try every question on your own first. Mark your work in a different colour, and write down the step where you went wrong.

Predict 1

1 • Pressure in solids

The stiletto hurts far more. Pressure, not weight, decides how much a surface gets dented.

$$\text{Stiletto: } p = 600 \text{ N} \div (1 \times 10^{-4} \text{ m}^2) = \mathbf{6\,000\,000 \text{ Pa.}}$$

$$\text{Elephant: } p = 50\,000 \text{ N} \div (4 \times 0.1 \text{ m}^2) = \mathbf{125\,000 \text{ Pa.}}$$

The heel presses about **48 times** harder on the floor, even though the elephant is over 80 times heavier.

Worked example 1

- 1 Greatest pressure needs the smallest face: 10 cm × 5 cm.

Convert to metres before multiplying.

$$\mathbf{A = 0.10 \text{ m} \times 0.050 \text{ m} = 0.0050 \text{ m}^2}$$

- 2 Apply $p = F / A$.

$$\mathbf{p_{\max} = 20 \text{ N} \div 0.0050 \text{ m}^2 = 4000 \text{ Pa}}$$

- 3 Least pressure uses the largest face: 20 cm × 10 cm.

Same force, 4 × the area, so $\frac{1}{4}$ of the pressure.

$$\mathbf{p_{\min} = 20 \text{ N} \div (0.20 \text{ m} \times 0.10 \text{ m}) = 20 \div 0.020 = 1000 \text{ Pa}}$$

Answer: **4000 Pa (standing on end) and 1000 Pa (lying flat)**

Q1

- 1 Convert the area.

$$\mathbf{A = 150 \text{ cm}^2 = 150 \times 10^{-4} \text{ m}^2 = 0.015 \text{ m}^2}$$

- 2 Apply $p = F / A$.

$$\mathbf{p = 600 \div 0.015 = 40\,000 \text{ Pa} = 40 \text{ kPa}}$$

Q2

(a)

- 1 Weight and total area.
 $F = mg = 3000 \times 10 = 30\,000\text{ N}$; $A = 4 \times 0.15 = 0.60\text{ m}^2$

- 2 Apply $p = F / A$.
 $p = 30\,000 \div 0.60 = 50\,000\text{ Pa (50 kPa)}$

(b)

Wide tyres give a **larger area** for the same weight, so the **pressure on the ground is lower** and the tractor does not sink into the soft mud.

2 • Pressure in liquids

Worked example
2

- 1 Pressure due to the seawater.
 $p = h\rho g = 20 \times 1030 \times 10 = 206\,000\text{ Pa}$

Formula stated and substituted. Usually worth a mark each.

- 2 The question says total, so add the air above the sea.
 $p_{\text{total}} = 206\,000 + 100\,000 = 306\,000\text{ Pa}$

Stopping at 206 000 Pa is the classic slip.

Answer: **$3.06 \times 10^5\text{ Pa (306 kPa)}$**

Error hunt 1

L2 is wrong. The depth was left in centimetres.

$h = 50\text{ cm} = 0.50\text{ m}$, so $p = 0.50 \times 1000 \times 10 = 5000\text{ Pa}$.

500 000 Pa would be the pressure 50 **m** down. Sanity check: 50 cm of water is nowhere near five atmospheres.

Q3

(a)

$$p = h\rho g = 3.0 \times 1000 \times 10 = 30\,000\text{ Pa}$$

(b)

$$F = pA = 30\,000 \times 2.0 = 60\,000\text{ N}$$

(c)

The bottom tap is **deeper**, so the **water pressure there is greater** and it pushes the water out faster.

Q4

No. Pressure due to a liquid depends only on depth and density: $p = h\rho g = 0.12 \times 1000 \times 10 = 1200\text{ Pa}$ for **both** bases.

The bowl's base has a **larger area**, so the total **force** on its base ($F = pA$) is bigger.

Q5

Water pressure **increases with depth**, so the water pushes hardest on the **bottom** of the dam. The wall must be thickest there to withstand the greatest pressure.

Worked example 3

3 • Hydraulic systems

- 1 (a) Pressure made by the effort. Convert the area first.

$$p = F_1 / A_1 = 50 \div (4.0 \times 10^{-4}) = 125\,000 \text{ Pa}$$

- 2 (b) The same pressure acts on the large piston.

$$F_2 = p \times A_2 = 125\,000 \times (200 \times 10^{-4}) = 2500 \text{ N}$$

Check with the ratio: $200 \div 4 = 50$, and $50 \times 50 \text{ N} = 2500 \text{ N}$.

- 3 (c) Volume of oil moved is the same on both sides.

$50 \times$ the force, $1/50$ of the distance.

$$4.0 \times 10 = 200 \times d_2, \text{ so } d_2 = 0.20 \text{ cm}$$

Answer: (a) $1.25 \times 10^5 \text{ Pa}$ (b) 2500 N (c) 0.20 cm

Error hunt 2

L2 is wrong. The areas are swapped: the effort acts on the **small** 2 cm^2 piston.

$$20 / 2 = F_2 / 50, \text{ so } F_2 = 500 \text{ N.}$$

Sanity check: a hydraulic press should make the force **bigger**, not smaller.

Q6

(a)

The **pressure** is transmitted **equally** through the brake fluid. The wheel pistons have a **larger area**, so $F = pA$ gives a larger force.

(b)

Air is **compressible**. Pressing the pedal squashes the bubble instead of transmitting the pressure to the wheels, so the brakes feel spongy and **the braking force is much smaller**.

4 • Atmospheric pressure and the barometer

Activity 1

Atmospheric pressure (about 100 kPa) pushes up on the card from below.

The water in the cup is only about 10 cm deep, so it pushes down on the card with a pressure of about $h\rho g = 0.10 \times 1000 \times 10 = 1000 \text{ Pa}$.

The upward push from the air is about 100 times bigger, so the card is held in place.

Worked example 4

- 1 Convert h to metres.

$$h = 60 \text{ cm} = 0.60 \text{ m}$$

- 2 Apply $p = h\rho g$.

$$p = 0.60 \times 13\,600 \times 10 = 81\,600 \text{ Pa}$$

Lower than at sea level, as expected up a mountain.

Answer: $8.16 \times 10^4 \text{ Pa}$

Q7 • MCQ

C

Only atmospheric pressure decides the vertical height. Tilting the tube means a longer length of tube is needed to reach the same vertical height.

Worked example
5

- 1 The open side is higher, so the gas pressure is above atmospheric.

Decide plus or minus first. It is worth a mark on its own.

$$p_{\text{gas}} = p_{\text{atm}} + h\rho g$$

- 2 Pressure difference from the level difference.

$$h\rho g = 0.25 \times 1000 \times 10 = 2500 \text{ Pa}$$

- 3 Add atmospheric pressure.

$$p_{\text{gas}} = 100\,000 + 2500 = 102\,500 \text{ Pa}$$

Answer: **$1.025 \times 10^5 \text{ Pa}$**

Q8 • MCQ

B

The pressure difference $h\rho g$ does not depend on area, so for the same gas pressure h is the same.

Q9

(a)

Below. The atmosphere pushes the mercury further down on the open side than the gas does on its side.

(b)

- 1 Subtract the level difference.

$$p_{\text{gas}} = 76 - 4.0 = 72 \text{ cm Hg}$$

- 2 Convert with $p = h\rho g$.

$$p = 0.72 \times 13\,600 \times 10 \approx 9.8 \times 10^4 \text{ Pa}$$

6 • Pull it together

Q10 • MCQ

B

Pressure = force ÷ area, so its unit is $\text{N} \div \text{m}^2$.

Q11 • MCQ

B

Depth is the same, so $p = h\rho g$ depends only on density. Water is denser, so the pressure is greater in X. The width does not matter.